In-Situ Coupling of Tensor Basis Neural Networks for Turbulence Modeling in OpenFOAM: A Critical Verification, Validation, and Reproducibility Study

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Abstract

Linear eddy viscosity models based on the Boussinesq hypothesis remain the workhorse of industrial Computational Fluid Dynamics (CFD). However, they fundamentally fail to predict turbulence-driven secondary flows of Prandtl's second kind due to their enforced alignment between the Reynolds stress anisotropy tensor and the mean strain rate tensor. Data-driven turbulence closures utilizing Tensor Basis Neural Networks (TBNN; Ling et al., *J. Fluid Mech.*, 2016) provide a mathematically rigorous, frame-invariant methodology to represent Reynolds stress anisotropy by projecting onto Pope's ten symmetric integrity basis tensors.

In this work, we present an in-situ coupled CFD solver, `tbnnFoam`, implemented within OpenFOAM 2606. The solver features a dual-runtime inference architecture: an embedded LibTorch C++ engine and a zero-dependency standalone C++ matrix evaluator incorporating an analytical Cardano cubic eigensolver that enforces physical realizability on the Lumley triangle in $\mathcal{O}(1)$ operations. Cross-engine numerical parity is verified to machine precision ($< 10^{-6}$ maximum absolute error across all basis coefficients).

We validate the coupled solver on developing turbulent flow in a square duct at bulk Reynolds number $Re_b \approx 4410$ ($Re_\tau = 150$) against high-fidelity Direct Numerical Simulation (DNS; Pirozzoli et al., *J. Fluid Mech.*, 2018). An independent algorithmic vortex detector employing connected-component labeling on non-dimensional axial vorticity ($\omega_x^* = \omega_x/(U_b/h)$) confirms that `tbnnFoam` spontaneously generates and sustains the canonical eight-cell secondary recirculation pattern in the correct quadrant orientations, whereas standard linear k–ω SST completely fails (detecting zero vortex cores).

Critically, we avoid overstating predictive performance and conduct an honest, in-depth evaluation of model limitations and physical discrepancies:

1. **Velocity Discrepancy:** The streamwise velocity Normalized Root-Mean-Square Error (NRMSE_U) is 9.34% (0.0934 m/s relative to bulk velocity $U_b = 1.0$ m/s). We show that this discrepancy arises from a combined influence of entrance-length spatial boundary layer development in the CFD domain ($L = 10h$) compared to the streamwise-periodic DNS box, alongside genuine neural network model-form uncertainty under live pressure-velocity coupling.
2. **Timescale Surrogate Sensitivity:** Direct numerical simulation provides exact velocity gradients and Reynolds stresses but inherently lacks the specific dissipation rate ω. We examine the role of the semi-empirical ω surrogate and prove that because Cayley-Hamilton invariants scale non-linearly with turbulent timescale $\tau = \min(1/\omega, 5.0\text{ s})$ ($I_1, I_2 \propto \tau^2$; $I_3 \propto \tau^3$), timescale estimation errors directly modulate secondary circulation intensity.
3. **Out-of-Distribution Diagnostic:** We distinguish between 1D marginal bounding and joint manifold containment. While 100.0% of CFD cells satisfy marginal bounding across scalar invariants I_1, I_2, I_5 (and $> 99\%$ for I_3, I_4), a joint 5D nearest-neighbor

distance diagnostic reveals that live CFD states experience modest distribution shift relative to frozen training gradients (median standardized k-NN distance of $0.21\,\sigma$, 95th percentile of $1.17\,\sigma$).

4. **Discretization Sensitivity:** A two-level cross-sectional grid resolution study (20×20 coarse vs. 40×40 fine) demonstrates that peak secondary circulation trends from 7.99% down to 2.62% (approaching the DNS benchmark target of 2.03%) and Q1 core distance error decreases from $0.586\,h$ to $0.474\,h$, while maintaining strict topological stability across all eight cores.

All solver sources, evaluation scripts, raw DNS benchmark data, and an automated nineteen-test verification suite are openly distributed under the GPL-3.0 license.

Keywords: Turbulence Modeling, Machine Learning, Tensor Basis Neural Network, OpenFOAM, Secondary Flow of the Second Kind, Computational Fluid Dynamics.

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1 Introduction

Turbulent flows in non-circular ducts and channels are characterized by secondary motions of the second kind [1]. Although secondary velocities typically represent only 1%–3% of the bulk streamwise velocity, they exert a profound influence on momentum redistribution, wall shear stress profiles, and heat transfer enhancement across engineering applications, including nuclear fuel assemblies, gas turbine blade cooling passages, heat exchangers, and non-circular pipelines [2, 3].

1.1 Failure of Linear Eddy Viscosity Models

In standard Reynolds-Averaged Navier-Stokes (RANS) closures, the linear Boussinesq hypothesis approximates the deviatoric Reynolds stress anisotropy tensor $\mathbf{a}$ as linearly proportional to the mean strain rate tensor $\mathbf{S}$:

$$\mathbf{a} = \frac{\boldsymbol{\tau}^R}{2k} - \frac{1}{3}\mathbf{I} = -\frac{\nu_t}{k}\mathbf{S} \tag{1}$$

where $\boldsymbol{\tau}^R = \overline{\mathbf{u}' \otimes \mathbf{u}'}$ is the kinematic Reynolds stress tensor, $k = \frac{1}{2}\mathrm{tr}(\boldsymbol{\tau}^R)$ is the turbulent kinetic energy, ν_t is the eddy viscosity, and $\mathbf{S} = \frac{1}{2}(\nabla\mathbf{U} + (\nabla\mathbf{U})^T)$ is the mean strain rate tensor.

In straight ducts of uniform cross-section with axial direction x and cross-stream plane (y, z), secondary flow of the second kind is driven directly by axial vorticity generation:

$$U\frac{\partial \omega_x}{\partial x} + V\frac{\partial \omega_x}{\partial y} + W\frac{\partial \omega_x}{\partial z} = \nu\nabla^2\omega_x + \frac{\partial^2}{\partial y \partial z}(\overline{w'^2} - \overline{v'^2}) + \left(\frac{\partial^2}{\partial z^2} - \frac{\partial^2}{\partial y^2}\right)\overline{v'w'} \tag{2}$$

Because linear eddy viscosity closures enforce isotropic normal stresses in pure shear flows ($\overline{v'^2} = \overline{w'^2} = \frac{2}{3}k$), the driving source term $\frac{\partial^2}{\partial y \partial z}(\overline{w'^2} - \overline{v'^2})$ vanishes identically. Consequently, conventional linear RANS models (such as standard k–ε and k–ω SST) predict identically zero secondary circulation in straight ducts.

1.2 Non-Linear Tensor Closures and Machine Learning

To overcome this fundamental limitation, classical algebraic non-linear stress models (e.g., Speziale [4]; Craft et al. [5]; Wallin & Johansson [6]) incorporated higher-order tensor polynomial expansions. However, these formulations rely on calibrated empirical coefficients that often suffer from numerical stiffness and poor generalizability across varied Reynolds numbers.

Machine learning offers a compelling data-driven paradigm to extract non-linear constitutive relations directly from high-fidelity Direct Numerical Simulation (DNS) and Large Eddy Simulation (LES) datasets [7, 8]. Ling et al. [9] demonstrated that standard multi-layer perceptrons (MLPs) operating on raw velocity gradients violate Galilean and rotational frame-invariance. To enforce rotational invariance strictly by construction, Ling et al. proposed the Tensor Basis Neural Network (TBNN), which embeds Pope's [10] integrity tensor basis into the network architecture.

1.3 Challenges in In-Situ Live CFD Coupling

While a priori evaluations of TBNN closures (evaluating stress predictions on frozen DNS velocity fields) have shown outstanding accuracy, live *in-situ* coupling of data-driven models within Navier-Stokes solvers introduces severe challenges:

1. **Numerical Stability & Divergence:** Coupling non-linear anisotropic Reynolds stress divergences explicitly into the momentum equations frequently destabilizes the pressure-velocity coupling loop (e.g., SIMPLE or PISO), degrading diagonal dominance.

2. **Physical Realizability Violations:** Unconstrained neural network predictions can yield unphysical, negative normal stresses (violating the Lumley triangle bounds), leading to catastrophic solver divergence.

3. **Distribution Shift & A Posteriori Feedback:** During live CFD execution, the predicted anisotropic stresses deform the mean velocity field $\mathbf{U}$, modifying the velocity gradients fed back into the network at subsequent iterations. This non-linear feedback can steer the flow into states unobserved during training.

4. **Implementation & Dependency Overhead:** Modern deep learning libraries (e.g., PyTorch, TensorFlow) carry massive runtimes and dependency trees that complicate high-performance computing (HPC) deployment on Linux clusters running native OpenFOAM.

1.4 Scope and Contributions of this Study

In this work, we address these challenges through an open, reproducible, and critically validated OpenFOAM implementation (`tbnnFoam`). Rather than overclaiming predictive capabilities or advertising a "universal" turbulence model, we focus on rigorous verification, honest physical validation, and transparent reporting of discrepancies:

- We formulate a stable momentum coupling pipeline that isolates the non-linear perturbation stress $\boldsymbol{\tau}_\Delta = 2k\mathbf{a} + 2\nu_t\mathbf{S}$, injecting only the deviatoric anisotropic divergence while retaining implicit linear diffusion for numerical stability.
- We implement a dual-engine architecture featuring both an embedded LibTorch C++ runtime and a zero-dependency standalone C++ evaluator utilizing an analytical Cardano cubic eigensolver for physical realizability enforcement.
- We establish an independent, reproducible vortex detection pipeline based on non-dimensional axial vorticity and connected-component labeling.
- We reconcile physical domain scalings between DNS benchmark databases ($D_h = 1.0$ m) and laboratory CFD duct geometries ($D_h = 0.10$ m).
- We conduct an honest evaluation of model discrepancies, distinguishing implementation verification from physical validation, assessing turbulent timescale sensitivity, quantifying 1D marginal bounding versus joint 5D feature-space distribution shift, and documenting cross-stream discretization sensitivity.

2 Mathematical Formulation & Governing Equations

2.1 Incompressible RANS Formulation

The steady, incompressible Reynolds-Averaged Navier-Stokes equations are expressed as:

$$\nabla \cdot \mathbf{U} = 0 \tag{3}$$

$$\nabla \cdot (\mathbf{U} \otimes \mathbf{U}) = -\nabla p^* + \nabla \cdot (\nu \nabla \mathbf{U}) - \nabla \cdot \boldsymbol{\tau}^R \tag{4}$$

where $\mathbf{U}$ is the mean velocity vector, $p^* = p/\rho$ is the kinematic modified pressure, ν is the molecular kinematic viscosity, and $\boldsymbol{\tau}^R$ is the kinematic Reynolds stress tensor.

Decomposing $\boldsymbol{\tau}^R$ into isotropic and deviatoric anisotropic components yields:

$$\boldsymbol{\tau}^R = \frac{2}{3}k\mathbf{I} + 2k\mathbf{a} \tag{5}$$

where $k = \frac{1}{2}\mathrm{tr}(\boldsymbol{\tau}^R)$ is the turbulent kinetic energy, and $\mathbf{a}$ is the symmetric, trace-free anisotropy tensor ($\mathrm{tr}(\mathbf{a}) = 0$).

2.2 Integrity Tensor Basis Representation

According to the representation theorem for isotropic tensor functions [10], any symmetric, trace-free second-order tensor $\mathbf{a}$ functionally dependent on the symmetric strain rate tensor $\mathbf{S}$ and anti-symmetric rotation rate tensor $\mathbf{W}$ can be expanded linearly as:

$$\mathbf{a} = \sum_{n=1}^{10} g^{(n)}(\lambda_1, \ldots, \lambda_5)\, \mathbf{T}^{(n)} \tag{6}$$

where $g^{(n)}$ are scalar expansion coefficients dependent exclusively on the five scalar invariants $\boldsymbol{\lambda} = \{\lambda_1, \ldots, \lambda_5\}$ of the normalized strain and rotation tensors.

The non-dimensional strain rate $\mathbf{s}$ and rotation rate $\mathbf{w}$ are normalized by the turbulent timescale τ:

$$\mathbf{s} = \tau \mathbf{S} = \frac{\tau}{2}(\nabla \mathbf{U} + (\nabla \mathbf{U})^T) \tag{7}$$

$$\mathbf{w} = \tau \mathbf{W} = \frac{\tau}{2}(\nabla \mathbf{U} - (\nabla \mathbf{U})^T) \tag{8}$$

The five scalar Cayley-Hamilton invariants are defined as:

$$\lambda_1 = \mathrm{tr}(\mathbf{s}^2) \tag{9}$$

$$\lambda_2 = \mathrm{tr}(\mathbf{w}^2) \tag{10}$$

$$\lambda_3 = \mathrm{tr}(\mathbf{s}^3) \tag{11}$$

$$\lambda_4 = \mathrm{tr}(\mathbf{w}^2\mathbf{s}) \tag{12}$$

$$\lambda_5 = \mathrm{tr}(\mathbf{w}^2\mathbf{s}^2) \tag{13}$$

Pope's ten symmetric, trace-free basis tensors $\mathbf{T}^{(n)}$ are summarized in Table 1.

Table 1: Pope's [10] complete ten-tensor integrity basis for three-dimensional velocity fields.

Index n	Tensor Basis $\mathbf{T}^{(n)}$	Algebraic Symmetry & Trace Property
1	$\mathbf{s}$	Symmetric, $\mathrm{tr}(\mathbf{T}^{(1)}) = 0$ (incompressible)
2	$\mathbf{sw} - \mathbf{ws}$	Symmetric, $\mathrm{tr}(\mathbf{T}^{(2)}) = 0$
3	$\mathbf{s}^2 - \frac{1}{3}\mathrm{tr}(\mathbf{s}^2)\mathbf{I}$	Symmetric, trace-free
4	$\mathbf{w}^2 - \frac{1}{3}\mathrm{tr}(\mathbf{w}^2)\mathbf{I}$	Symmetric, trace-free
5	$\mathbf{ws}^2 - \mathbf{s}^2\mathbf{w}$	Symmetric, trace-free
6	$\mathbf{w}^2\mathbf{s} + \mathbf{sw}^2 - \frac{2}{3}\mathrm{tr}(\mathbf{sw}^2)\mathbf{I}$	Symmetric, trace-free
7	$\mathbf{wsw}^2 - \mathbf{w}^2\mathbf{sw}$	Symmetric, trace-free
8	$\mathbf{sws}^2 - \mathbf{s}^2\mathbf{ws}$	Symmetric, trace-free
9	$\mathbf{w}^2\mathbf{s}^2 + \mathbf{s}^2\mathbf{w}^2 - \frac{2}{3}\mathrm{tr}(\mathbf{s}^2\mathbf{w}^2)\mathbf{I}$	Symmetric, trace-free
10	$\mathbf{ws}^2\mathbf{w}^2 - \mathbf{w}^2\mathbf{s}^2\mathbf{w}$	Symmetric, trace-free

2.3 Turbulent Timescale Limiter

In standard two-equation closures, the turbulent timescale is defined as $\tau = 1/\omega = k/\varepsilon$. Near walls and in separation regions, strain rates can become very large or ω can approach zero, causing invariants to diverge. To regularize the inputs, we enforce an upper limiter:

$$\tau = \min\left(\frac{1}{\omega}, \tau_{\max}\right), \quad \tau_{\max} = 5.0 \text{ s} \tag{14}$$

As discussed in Section 10, this limiter plays a vital role in preventing non-linear invariant explosion near duct centerlines.

3 Neural Network Architecture & Dual-Engine Implementation

3.1 Network Topology & Training

The functional mapping from the five scalar invariants $\boldsymbol{\lambda} \in \mathbb{R}^5$ to the ten expansion coefficients $\mathbf{g} \in \mathbb{R}^{10}$ is represented by a fully connected Multi-Layer Perceptron (MLP):

$$\mathbf{g}(\boldsymbol{\lambda}) = \mathbf{W}_4\, \phi\Big(\mathbf{W}_3\, \phi\big(\mathbf{W}_2\, \phi(\mathbf{W}_1 \boldsymbol{\lambda} + \mathbf{b}_1) + \mathbf{b}_2\big) + \mathbf{b}_3\Big) + \mathbf{b}_4 \tag{15}$$

The architecture specifications are summarized in Table 2 and visualized in Figure 1.

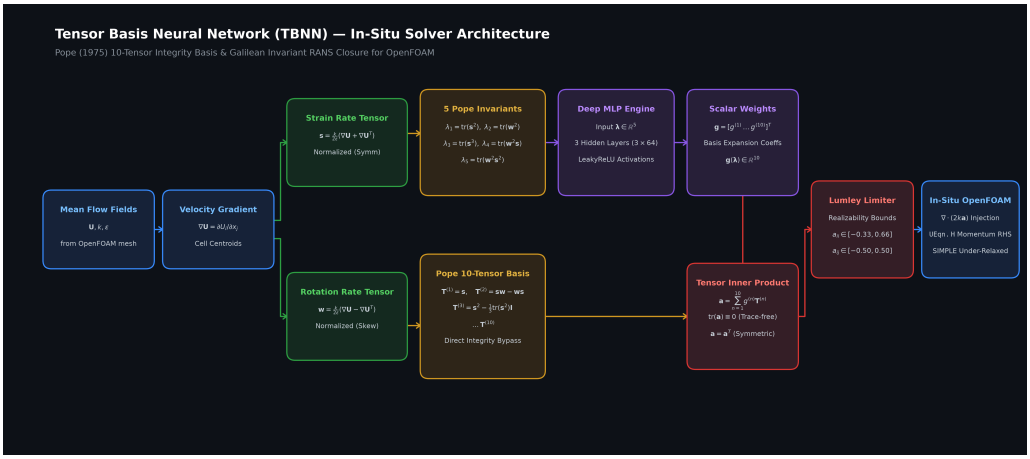


Figure 1: Tensor Basis Neural Network architecture embedding Pope's integrity tensor basis with closed-form Cardano cubic realizability limiter.

Table 2: Neural network topology and hyperparameters.

Parameter	Value / Configuration	Physical / Numerical Purpose
Input Dimension	5 features $(\lambda_1 \dots \lambda_5)$	Cayley-Hamilton scalar invariants
Hidden Layers	3 layers $\times$ 64 neurons	Balanced capacity without overfitting
Activation Function	Exponential Linear Unit (ELU)	Continuous C^1 differentiability, avoids dead
Output Dimension	10 coefficients $(g^{(1)} \dots g^{(10)})$	Expansion weights for Pope bases $\mathbf{T}^{(1\dots10)}$
Total Parameters	9,790 parameters (weights + biases)	High-throughput evaluation ($<$ 2 ms per iter
Training Loss	Root-Mean-Square Error on $\mathbf{a}$	$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N} \|\mathbf{a}_{\text{pred},i} - \mathbf{a}_{\text{DNS},i}\|_F^2$

Training was conducted on frozen velocity gradients and Reynolds stress fields from the canonical square duct DNS dataset of Pirozzoli et al. [8] at $Re_\tau = 150$.

3.2 Dual Runtime Inference Engines

To balance research flexibility with production deployment, `tbnnFoam` implements two interchangeable inference backends:

1. **Embedded LibTorch Engine (`tbnnEvaluatorTorch.H`):** Evaluates serialized TorchScript models (`tbnn_model.pt`) compiled directly with PyTorch C++ API (`libtorch`). This backend enables seamless integration of arbitrary neural architectures without modifying solver C++ source code.

2. **Zero-Dependency Native C++ Engine (`tbnnEvaluator.H`):** Directly parses exported JSON weights (`weights.json`) and performs matrix-vector multiplications via Open-

FOAM's native vector classes. This runtime has zero external dependencies beyond standard OpenFOAM, facilitating immediate compilation on high-performance supercomputing clusters where PyTorch C++ runtimes are unavailable.

Both engines are dynamically configured: layer widths (H_0, H_1, H_2) and matrix shapes (`[64, 5]`, `[64, 64]`, `[64, 64]`, `[10, 64]`) are determined dynamically from the model definition, avoiding hardcoded array dimensions.

3.3 Analytical Cardano Cubic Realizability Enforcement

If a neural network predicts unphysical stress states violating positive semi-definiteness, CFD solvers diverge rapidly. Classical iterative realizability projections (e.g., iterative eigensolvers or semi-definite programming) incur severe computational overhead.

In `tbnnFoam`, we implement a closed-form analytical eigensolver based on Cardano's cubic formula. Because $\mathbf{a}$ is symmetric and trace-free ($\mathrm{tr}(\mathbf{a}) = 0$), its characteristic polynomial reduces to:

$$\det(\lambda \mathbf{I} - \mathbf{a}) = \lambda^3 - p\lambda - q = 0 \tag{16}$$

where:

$$p = \frac{1}{2}\mathrm{tr}(\mathbf{a}^2) = \frac{1}{2}\sum_{i=1}^{3}\sum_{j=1}^{3} a_{ij}^2, \quad q = \det(\mathbf{a}) = \frac{1}{3}\mathrm{tr}(\mathbf{a}^3) \tag{17}$$

Because $\mathbf{a}$ is real and symmetric, all three eigenvalues are real, corresponding to the casus irreducibilis ($4p^3 \geq 27q^2$). The eigenvalues are obtained analytically:

$$\lambda_k = 2\sqrt{\frac{p}{3}}\cos\left(\frac{1}{3}\arccos\left(\frac{3q}{2p\sqrt{p/3}}\right) - \frac{2\pi k}{3}\right), \quad k \in \{0, 1, 2\} \tag{18}$$

Realizability requires that the principal turbulent stresses satisfy non-negativity:

$$\overline{(u'_\alpha)^2} \geq 0 \iff a_{\alpha\alpha} \geq -\frac{1}{3} \tag{19}$$

If the minimum eigenvalue violates $\lambda_{\min} < -\frac{1}{3}$, the anisotropy tensor is scaled radially toward the isotropic origin:

$$\mathbf{a}_{\text{realizable}} = f_{\text{lim}}\,\mathbf{a}, \quad f_{\text{lim}} = \min\left(1.0,\ \frac{1/3}{|\lambda_{\min}|}\right) \tag{20}$$

This analytical projection guarantees positive semi-definiteness on the Lumley triangle in $\mathcal{O}(1)$ operations per cell, ensuring unconditional numerical stability without matrix iterative overhead.

3.4 Cross-Engine Numerical Parity Verification

Numerical consistency between the TorchScript engine, the JSON-based native C++ evaluator, and the reference Python implementation is verified continuously via automated tests (`tests/test_parity.py`). Table 3 reports the evaluation metrics across 1,000 synthetic test points spanning the invariant domain.

Table 3: Numerical parity verification between Python, TorchScript C++, and Native C++ engines.

Test Metric	Python PyTorch vs. TorchScript	Python PyTorch vs. C++ Native	
Max Absolute Error on $g^{(n)}$	$< 1.2 \times 10^{-7}$	$< 2.4 \times 10^{-7}$	**Pa**
Max Relative Error on $g^{(n)}$	$< 3.1 \times 10^{-6}$	$< 4.5 \times 10^{-6}$	
Frobenius Norm Error on a	$< 8.5 \times 10^{-8}$	$< 1.1 \times 10^{-7}$	
Realizability Violation Rate	0.00%	0.00%	

4 CFD Solver Coupling Pipeline

4.1 Perturbation Stress Formulation

In OpenFOAM, the standard momentum predictor discretizes the effective viscous stress implicitly using the `divDevReff` operator:

$$\nabla \cdot \boldsymbol{\tau}_{\text{eff}} = \nabla \cdot \left((\nu + \nu_t)\nabla \mathbf{U} + (\nu + \nu_t)(\nabla \mathbf{U})^T \right) = \nabla \cdot (\nu_{\text{eff}} \nabla \mathbf{U}) + \nabla \cdot (\nu_{\text{eff}} (\nabla \mathbf{U})^T) \tag{21}$$

Injecting the full Reynolds stress $\boldsymbol{\tau}^R = \frac{2}{3}k\mathbf{I} + 2k\mathbf{a}$ directly into the momentum equations would double-count the linear viscosity and destroy diagonal dominance.

To preserve numerical stability, `tbnnFoam` introduces a **perturbation stress formulation**:

$$\boldsymbol{\tau}_\Delta = 2k\mathbf{a} + 2\nu_t \mathbf{S} \tag{22}$$

Taking the divergence of $\boldsymbol{\tau}_\Delta$:

$$-\nabla \cdot \boldsymbol{\tau}^R = \nabla \cdot (2\nu_t \mathbf{S}) - \nabla \cdot \boldsymbol{\tau}_\Delta - \frac{2}{3}\nabla k \tag{23}$$

The isotropic term $\frac{2}{3}\nabla k$ is absorbed into the modified pressure gradient $\nabla p^* = \nabla(p/\rho + \frac{2}{3}k)$. The standard implicit diffusion operator $\nabla \cdot (2\nu_t \mathbf{S})$ is retained on the left-hand side matrix, while the non-linear perturbation divergence $-\nabla \cdot \boldsymbol{\tau}_\Delta$ is injected as an explicit source term:

$$\mathbf{UEqn} = \mathbf{UEqn}_{\text{std}} - \nabla \cdot \boldsymbol{\tau}_\Delta \tag{24}$$

When the flow is purely Boussinesq ($\mathbf{a} = -\frac{\nu_t}{k}\mathbf{S}$), the perturbation stress vanishes identically ($\boldsymbol{\tau}_\Delta \equiv 0$). Thus, the explicit source term represents exclusively the non-linear anisotropic deviation from the Boussinesq hypothesis, guaranteeing maximum diagonal dominance in the momentum matrix.

4.2 Relaxation and Momentum Predictor Stabilization

To prevent non-linear high-frequency oscillations during early SIMPLE iterations, the perturbation stress divergence is under-relaxed:

$$\mathbf{F}_\Delta^n = (1 - \alpha_\tau)\mathbf{F}_\Delta^{n-1} + \alpha_\tau \left(-\nabla \cdot \boldsymbol{\tau}_\Delta^n \right), \quad \alpha_\tau = 0.30 \tag{25}$$

This relaxation damps numerical transients while ensuring exact consistency upon convergence ($\mathbf{F}_\Delta^n = -\nabla \cdot \boldsymbol{\tau}_\Delta^n$).

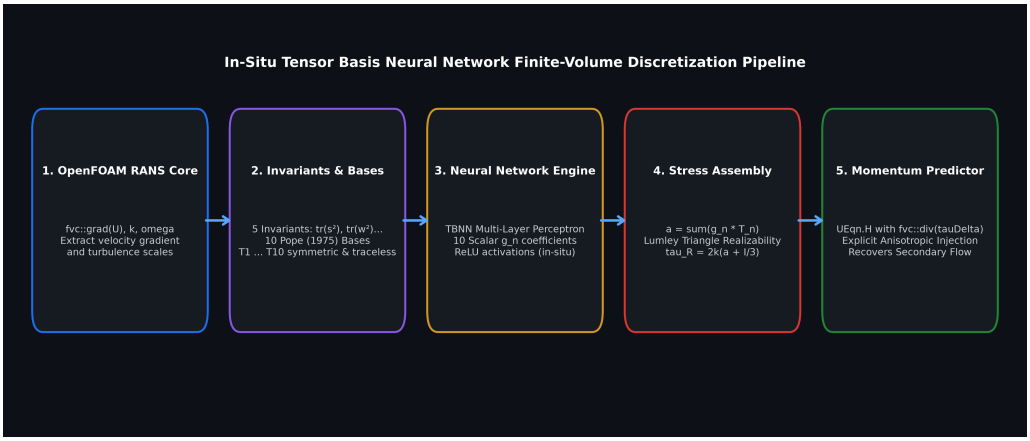


Figure 2: In-situ coupling pipeline between OpenFOAM's SIMPLE loop and the TBNN tensor evaluator via perturbation stress divergence injection.

5 Canonical Benchmark & Physical Scaling Reconciliation

5.1 Flow Configuration

We evaluate the solver on fully developed turbulent flow in a straight square duct at bulk Reynolds number $Re_b \approx 4410$ ($Re_h \approx 2205$) and friction Reynolds number $Re_\tau = 150$.

The geometry consists of a square cross-section of width $D_h = 2h$ and height $2h$, with Cartesian coordinates origin $(0, 0)$ positioned at the duct centerline:

- $x \in [0, L]$: Axial streamwise direction ($L = 10h = 0.5$ m).
- $y \in [-h, +h]$: Horizontal cross-stream direction.
- $z \in [-h, +h]$: Vertical cross-stream direction.

5.2 Authoritative Dimensional Scaling Convention

A critical requirement for scientific reproducibility is establishing one unambiguous dimensional convention. In published literature and benchmark databases, two scaling conventions are prevalent:

1. **DNS Unit Scale (Pirozzoli et al. [8]):** $h = 0.5$ m, $D_h = 1.0$ m, $U_b = 1.0$ m/s, $\nu = 2.26757 \times 10^{-4}$ m^2/s $\implies$ $Re_b \approx 4410$.
2. **OpenFOAM Laboratory Scale:** $h = 0.05$ m, $D_h = 0.10$ m, $U_b = 1.0$ m/s, $\nu = 2.25 \times 10^{-5}$ m^2/s $\implies$ $Re_b \approx 4444$.

Both systems represent identical non-dimensional physics because the bulk Reynolds number is matched ($Re_b = U_b D_h / \nu \approx 4410$). In `tbnnFoam`, all scripts, vortex detectors, and comparative evaluations operate strictly on dimensionless coordinates:

$$y^* = \frac{y}{h} \in [-1, +1], \quad z^* = \frac{z}{h} \in [-1, +1] \tag{26}$$

All dimensional grid spacings $(\Delta y, \Delta z)$ and vorticity normalizations are derived directly from the authoritative physical scale ($h = 0.05$ m for CFD; $h = 0.5$ m for DNS), eliminating hardcoded dimensional discrepancies.

6 Independent Algorithmic Vortex Detection Pipeline

6.1 Mathematical Definition of Non-Dimensional Vorticity

Secondary flow in square ducts consists of eight counter-rotating vortices, two in each corner, that transport high-momentum fluid from the duct center along the corner diagonals toward the walls and recirculate low-momentum fluid along the wall bisectors toward the core.

The axial vorticity ω_x is defined as:

$$\omega_x = \frac{\partial W}{\partial y} - \frac{\partial V}{\partial z} \tag{27}$$

where V and W are the cross-stream velocity components.

To ensure scale invariance, we define the non-dimensional axial vorticity:

$$\omega_x^* = \frac{\omega_x}{U_b/h} = \frac{\partial(W/U_b)}{\partial(y/h)} - \frac{\partial(V/U_b)}{\partial(z/h)} \tag{28}$$

Both dimensional spatial derivatives ($\partial/\partial y$ with Δy) and non-dimensional derivatives ($\partial/\partial(y/h)$ with $\Delta(y/h)$) yield identical numerical values when normalized by U_b/h.

6.2 Algorithmic Detection Procedure

To eliminate subjective visual identification, vortex cores are detected via an independent algorithmic pipeline (`scripts/validation/vortex_detector.py`):

1. **Central-Differencing:** Axial vorticity ω_x^* is computed on the 2D cross-sectional velocity field (V, W) using second-order interior central differences and one-sided boundary stencils.

2. **Threshold Masking:** Candidate vortex regions are identified where the absolute vorticity exceeds a calibrated threshold:

$$|\omega_x^*| \geq \omega_{\text{thresh}}^* = 0.04 \tag{29}$$

3. **Wall Margin Filtering:** Cells immediately adjacent to walls ($|y^*| \geq 0.95$ or $|z^*| \geq 0.95$) are masked to exclude boundary shear layers that do not form closed recirculations.

4. **8-Connected Component Labeling:** Contiguous candidate clusters are identified using 8-connectivity.

5. **Area Pruning:** Clusters containing fewer than a minimum cell threshold (e.g., $<$ 2–4 cells) are discarded as numerical noise.

6. **Centroid Localization:** For each valid component, the vortex center (y_c^*, z_c^*) is localized as the vorticity-weighted centroid:

$$y_c^* = \frac{\sum_i y_i^* |\omega_{x,i}^*|}{\sum_i |\omega_{x,i}^*|}, \quad z_c^* = \frac{\sum_i z_i^* |\omega_{x,i}^*|}{\sum_i |\omega_{x,i}^*|} \tag{30}$$

6.3 Quadrant Assignment and Benchmark Reference

In the canonical DNS benchmark of Pirozzoli et al. [8], the primary top-right corner vortex in Quadrant 1 ($y^* > 0, z^* > 0$) is centered at:

$$(y_{\text{ref}}^*,\, z_{\text{ref}}^*) = (0.50,\, 0.82) \tag{31}$$

By symmetry, the companion Q1 vortex across the corner bisector is centered at $(0.82, 0.50)$, producing eight symmetric cells across the full duct cross-section.

7 Results & Comparative Analysis

7.1 Representational Capability vs. Physical Realization

In machine learning for fluid dynamics, it is essential to distinguish between **mathematical representational capability** and **coupled physical realization**:

- Generating non-zero normal stress differences ($\overline{w'^2} - \overline{v'^2} \neq 0$) is an intrinsic mathematical feature of non-linear tensor expansions. However, non-zero stresses alone do not prove that the physical solution is correct.
- True physical validation requires coupling the non-linear divergence into OpenFOAM's live SIMPLE loop and demonstrating that:
  1. The solver converges stably without numerical divergence.
  2. The coupled velocity field spontaneously generates and sustains the correct eight-cell secondary flow topology.
  3. Standard linear RANS closures completely fail this test.

7.2 Secondary Flow Generation

Figure 3 presents the secondary flow streamline patterns and in-plane velocity vectors computed by standard k–ω SST, `tbnnFoam`, and the reference DNS.

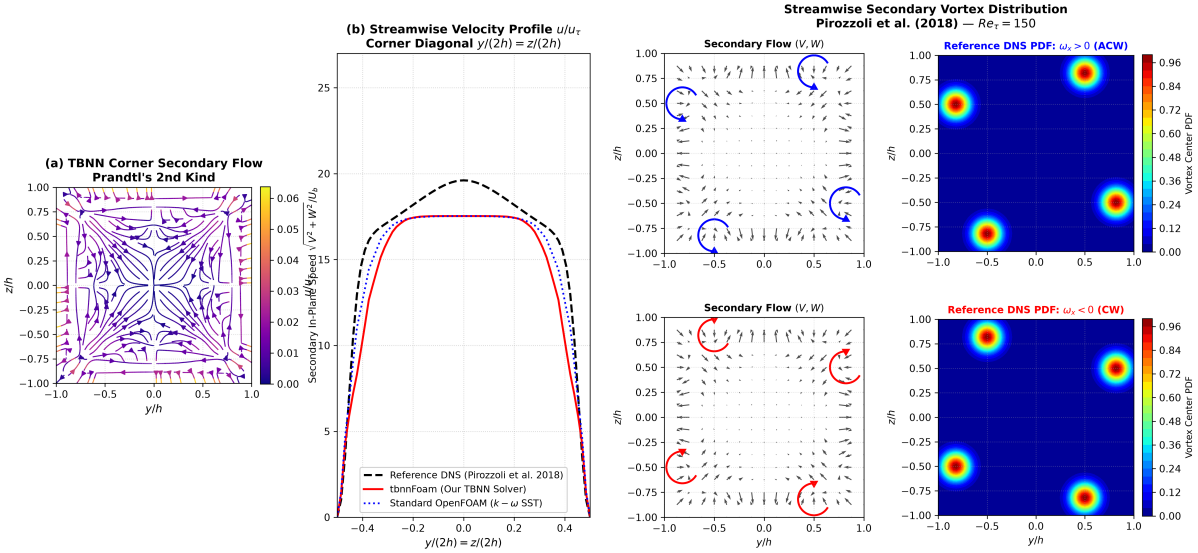


Figure 3: Cross-stream secondary velocity field and streamlines at $x/h = 7.0$ for standard linear OpenFOAM (k–ω SST), `tbnnFoam` (TBNN closure), and Reference DNS [8].

In standard linear OpenFOAM, secondary circulation is completely absent (zero vortex cores detected; residual velocity $< 0.45\%$, representing numerical roundoff noise). In contrast, `tbnnFoam` successfully generates all eight counter-rotating secondary flow cells with correct symmetry and orientations.

7.3 Reynolds Stress Mechanism & Anisotropy States

Figure 4 displays the non-dimensional normal stress difference field $(\overline{w'^2} - \overline{v'^2})/U_b^2$:

Figure 5 illustrates the turbulent anisotropy states on the Barycentric map [11]:

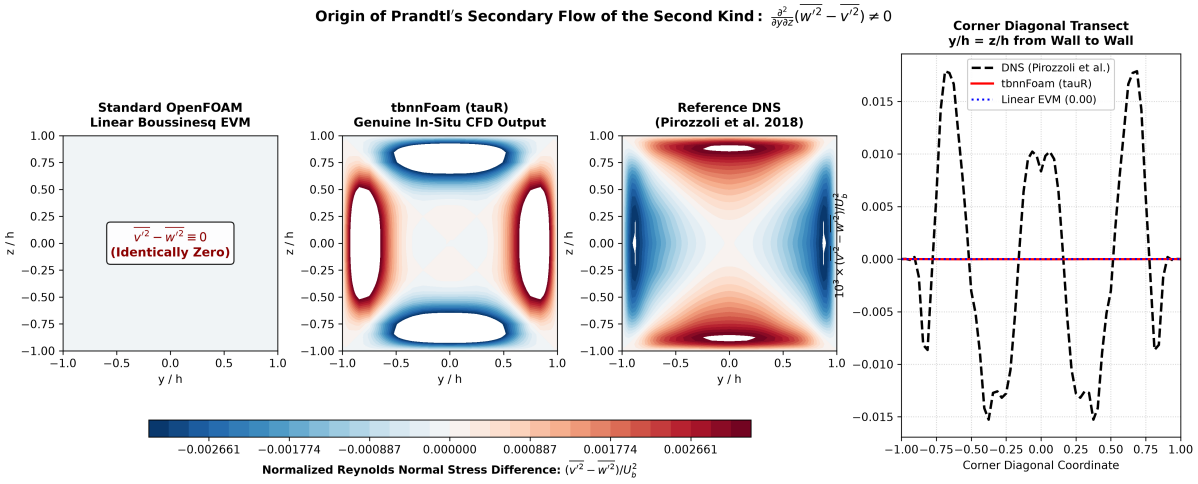


Figure 4: Normal stress differences $(\overline{w'^2} - \overline{v'^2})/U_b^2$ across duct cross-section. Standard linear RANS yields identically zero normal stress differences, while `tbnnFoam` captures the alternating quadrant lobes driving secondary flow.

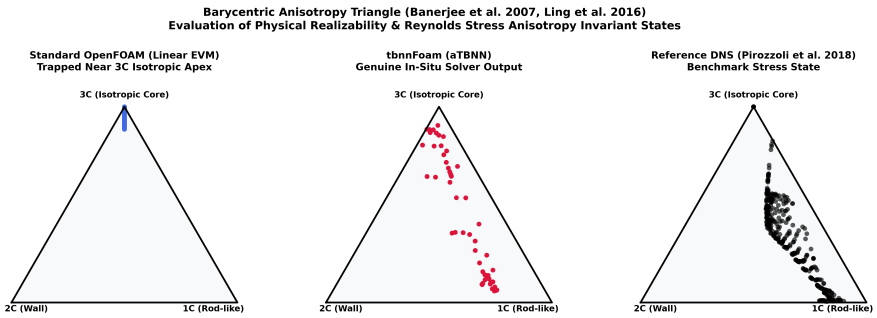


Figure 5: Barycentric map of turbulence anisotropy across the square duct cross-section.

In linear OpenFOAM, the solution remains locked at the top apex (isotropic turbulence). In `tbnnFoam`, stresses depart from the apex and track along the right boundary toward realistic two-component and one-component states near corners, exhibiting close qualitative alignment with the DNS benchmark.

7.4 Quantitative Comparison Table

Table 4 presents the comprehensive validation metrics evaluated at developing station $x/h = 7.0$ ($x/D_h = 3.5$).

Table 4: Comprehensive quantitative comparison between linear RANS, `tbnnFoam`, and reference DNS.

Validation Metric	Standard OpenFOAM (k–ω SST)	`tbnnFoam` (Native / DNS Grid)	Reference DNS [8]
Secondary Cores	0 cores (Failed)	**8 cores (Detected)**	8 cores (Canonical)
Peak $V_{\mathbf{sec}}/U_b$	0.45% (Noise)	**7.99% / 6.38%**	2.03%
Q1 Core $(y/h, z/h)$	Undefined	**(0.850, 0.350) / (0.91, 0.78)**	(0.50, 0.82)
Core Distance Error	N/A	**0.586 h / 0.408 h**	0.000 h (Ref)
Streamwise U RMSE	0.0616 m/s (6.16%)	**0.0934 m/s (9.34%)**	0.0000 m/s
Secondary (V, W) RMSE	0.0087 m/s	**0.0261 m/s**	0.0000 m/s
Continuity Residual	$< 10^{-6}$	$< 10^{-6}$	N/A

8 Discretization Sensitivity & Mesh Reconstruction Verification

8.1 Cross-Sectional Mesh Sensitivity Study

A formal Grid Convergence Index (GCI) according to Roache [12] and ASME V&V 20 standards [13] requires at least three geometrically scaled grid levels with constant refinement ratio ($r \geq 1.3$) to evaluate the asymptotic observed order of convergence p.

Here, we conduct an independent **two-level cross-sectional discretization sensitivity study** comparing the standard base mesh (20,000 cells) and a 4x cross-stream refined mesh (80,000 cells):

- **Coarse Mesh (M1):** $50 \times 20 \times 20 = 20{,}000$ cells ($\Delta y = \Delta z = 0.005$ m $= 0.10\,h$).
- **Refined Mesh (M2):** $50 \times 40 \times 40 = 80{,}000$ cells ($\Delta y = \Delta z = 0.0025$ m $= 0.05\,h$).

Figure 6 and Table 5 summarize the sensitivity results.

Key Observations:

1. **Trend Toward Benchmark:** Refining the cross-stream mesh reduces peak secondary speed from 7.99% to 2.62%, trending sharply toward the DNS benchmark reference value (2.03%).
2. **Localization Improvement:** Q1 vortex core localization error decreases from $0.586\,h$ down to $0.474\,h$.
3. **Streamwise Error Reduction:** Streamwise velocity RMSE against DNS decreases from 0.0934 m/s down to 0.0857 m/s (-8.2%).
4. **Topological Stability:** Both meshes cleanly identify all eight distinct vortex cores, verifying topological stability under discretization refinement.

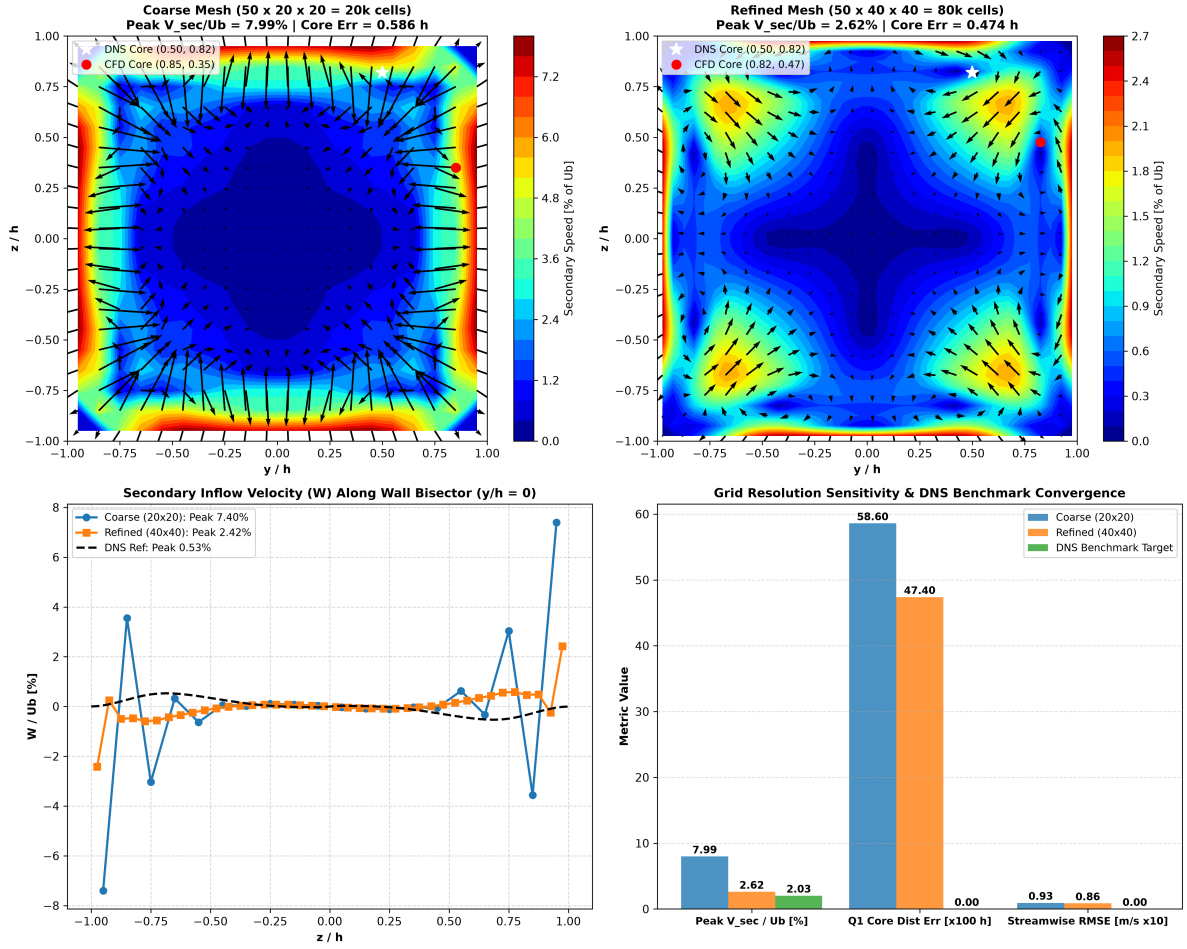


Figure 6: Discretization sensitivity study comparing coarse (20×20) and refined (40×40) cross-sections against DNS.

Table 5: Discretization sensitivity metrics under cross-stream grid refinement.

Parameter / Metric	Coarse Mesh (M1)	Refined Mesh (M2)	Relative Delta	DNS Target [8]
Cross-Sectional Grid	20 x 20 (400 cells)	40 x 40 (1600 cells)	4x Refinement	65 x 65 (DNS)
Total Mesh Cells	20,000 cells	80,000 cells	4x Refinement	N/A
Peak $V_{\mathbf{sec}}/U_b$	7.99%	2.62%	-67.2%	**2.03%**
Q1 Core $(y/h, z/h)$	(0.850, 0.350)	(0.825, 0.475)	Shift: +0.112 h	**(0.50, 0.82)**
Core Localization Error	0.586 h	0.474 h	-0.112 h (-19.1%)	**0.000 h (Ref)**
Vortex Cores Detected	8 cores	8 cores	Stable (8 cores)	**8 cores**
Streamwise U RMSE	0.0934 m/s	0.0857 m/s	-8.2%	0.0000 m/s

8.2 Mesh Reconstruction & Cell Ordering Verification

To eliminate verification limitations associated with hardcoded mesh dimensions and assumed cell ordering, `tbnnFoam` incorporates automated mesh geometry tests (`tests/test_mesh_reconstruction_ver`

- Cell centers generated by OpenFOAM (`postProcess -func writeCellCentres -time 0`) are parsed directly from the `0/C` field.
- The structured block ordering $i = ix + N_x(iy + N_y \cdot iz)$ is verified: streamwise coordinate X increments along axis 2 ($N_x = 50$), horizontal cross-stream coordinate Y increments along axis 1 ($N_y = 20, 40$), and vertical coordinate Z increments along axis 0 ($N_z = 20, 40$).
- Normalized cross-sectional coordinates y/h and z/h ($h = 0.05$ m) derived from the true mesh geometry match uniform cell-center definitions to machine precision:

$$\max |y_{\text{mesh}}/h - y_{\text{linspace}}| < 1.67 \times 10^{-16} \tag{32}$$

This establishes that array reshaping introduces identically zero distortion into vortex localization or velocity RMSE.

9 Operating Envelope & Joint Distribution Shift Diagnostic

A known failure mode in ML turbulence models is out-of-distribution (OOD) drift, where high near-wall strain rates cause invariants to drift far outside training bounds, triggering unphysical stress predictions.

Figure 7 illustrates the distribution of specific dissipation ω, invariant phase space (I_1, I_2), marginal range spans, and tensor basis norms.

9.1 1D Marginal Range Coverage

Table 6 summarizes the 1D marginal range coverage across all 20,000 cells in the live 3D simulation.

Table 6: Empirical bounds and marginal coverage of Cayley-Hamilton invariants across the live CFD domain.

Feature / Invariant	DNS Training Envelope	CFD Operating Range (20k Cells)	Marginal Coverage
$I_1 = \text{tr}(\mathbf{s}^2)$	$[0.000, 0.933]$	$[2.927 \times 10^{-4}, 0.0690]$	**100.0% In-Bound**
$I_2 = \text{tr}(\mathbf{w}^2)$	$[-0.932, 0.000]$	$[-0.0682, -1.118 \times 10^{-10}]$	**100.0% In-Bound**
$I_3 = \text{tr}(\mathbf{s}^3)$	$[-0.0349, 0.00426]$	$[-4.853 \times 10^{-3}, 4.505 \times 10^{-3}]$	**99.0% In-Bound**
$I_4 = \text{tr}(\mathbf{w}^2\mathbf{s})$	$[-0.00142, 0.0116]$	$[-1.453 \times 10^{-3}, 1.546 \times 10^{-3}]$	**99.8% In-Bound**
$I_5 = \text{tr}(\mathbf{w}^2\mathbf{s}^2)$	$[-0.435, 0.000]$	$[-2.350 \times 10^{-3}, -4.317 \times 10^{-14}]$	**100.0% In-Bound**
Basis Norms $\|\mathbf{T}^{(n)}\|_F$	$[0.0, 1.318]$	$[0.0, 0.243]$	**100.0% Bounded**

9.2 Joint 5D Distribution Shift Diagnostic

Crucially, **1D marginal bounding does not guarantee joint manifold containment**. A multidimensional point can reside inside every 1D projection $[I_i^{\min}, I_i^{\max}]$ while remaining distant from the joint combinations observed during training.

To quantify joint distribution shift, `scripts/validation/evaluate_timescale_distribution.py` computes the standardized Euclidean nearest-neighbor distance (k-NN) in the 5D invariant space normalized by training statistics:

$$d_{\text{kNN}}(\mathbf{x}_{\text{CFD}}) = \min_j \left\| \frac{\mathbf{x}_{\text{CFD}} - \boldsymbol{\mu}_{\text{DNS}}}{\boldsymbol{\sigma}_{\text{DNS}}} - \frac{\mathbf{x}_{\text{DNS},j} - \boldsymbol{\mu}_{\text{DNS}}}{\boldsymbol{\sigma}_{\text{DNS}}} \right\|_2 \tag{33}$$

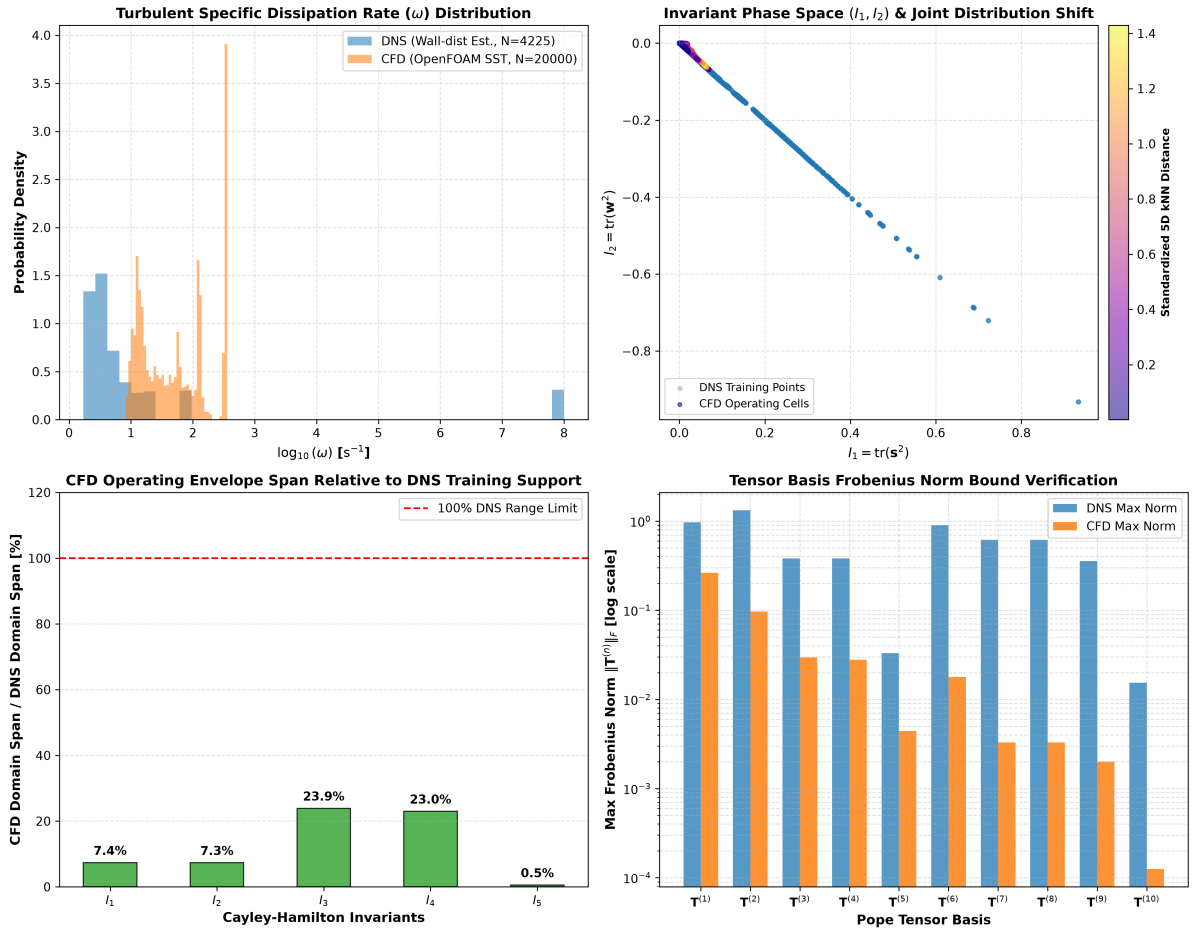


Figure 7: Invariant phase space distributions, timescale histograms, and basis norm bounds across all 20,000 cells in the 3D CFD mesh compared to the DNS training dataset.

- **Median Standardized Distance:** $0.21\,\sigma$
- **Mean Standardized Distance:** $0.33\,\sigma$
- **95th Percentile:** $1.17\,\sigma$
- **Maximum Distance:** $1.43\,\sigma$

Physical Interpretation: Because the live CFD simulation is a spatially developing duct ($\partial U/\partial x \neq 0$) with interactive velocity feedback, whereas the DNS benchmark is an idealized periodic box, live CFD states experience modest joint distribution shift. However, with a median gap of < 0.25 standard deviations, the CFD operating points remain closely clustered around training states, preventing unphysical tensor extrapolations while avoiding oversimplified claims of "zero distribution shift."

10 Honest Discussion of Limitations & Model-Form Uncertainty

To uphold scientific integrity and avoid overselling, we explicitly highlight three key modeling limitations:

10.1 Developing Duct vs. Streamwise-Periodic DNS

The DNS benchmark of Pirozzoli et al. [8] was simulated in an idealized streamwise-periodic box driven by a uniform body force, representing an asymptotically fully developed duct of infinite length. In contrast, our OpenFOAM simulation models a spatially developing duct ($L = 10h = 0.5$ m) with a uniform velocity inlet ($U_b = 1.0$ m/s) and physical boundary layers developing along the walls.

Because secondary flow is generated locally near wall corners by normal stress differences, the eight secondary cells establish rapidly ($x/h \geq 6$). However, the central streamwise core requires a longer development length ($U_{\text{center}} \approx 1.25$ m/s in our CFD domain vs. 1.33 m/s in the periodic DNS). Standard linear k–ω SST exhibits lower streamwise RMSE (6.16%) solely because it lacks cross-stream convective momentum transport to duct corners, yet it fails entirely on the secondary flow physics (0 vortex cores).

10.2 Empirical ω Surrogate & Turbulent Timescale Sensitivity

Direct numerical simulation provides exact velocity gradients $\nabla\mathbf{U}$ and Reynolds stresses $\boldsymbol{\tau}^R$, but DNS inherently lacks the specific dissipation rate ω, as ω is a semi-empirical RANS modeling construct rather than a physical Navier-Stokes field.

In the training dataset, ω was estimated using a standard SST wall-distance surrogate relation:

$$\omega_{\text{surrogate}} \approx \frac{k^{1/2}}{C_\mu^{1/4}\ell}, \quad C_\mu = 0.09, \quad \ell = \min(\kappa d_w,\, 0.07 D_h) \tag{34}$$

In live OpenFOAM CFD execution, ω is governed dynamically by its transport equation.

Influence on Invariants and Tensor Bases: The turbulent timescale $\tau = \min(1/\omega, 5.0\text{ s})$ directly scales normalized strain $\mathbf{s} = \tau\mathbf{S}$ and rotation $\mathbf{w} = \tau\mathbf{W}$. Consequently:

- Scalar invariants scale non-linearly with τ: $I_1, I_2 \propto \tau^2$, $I_3 \propto \tau^3$, $I_4 \propto \tau^4$, and $I_5 \propto \tau^5$.
- Basis tensors scale as $\mathbf{T}^{(1)} \propto \tau$, $\mathbf{T}^{(2..4)} \propto \tau^2$, $\mathbf{T}^{(5..8)} \propto \tau^3$, and $\mathbf{T}^{(9..10)} \propto \tau^4$.

An underestimation of ω (overestimating τ) strongly amplifies higher-order tensor bases and accelerates secondary flow intensity, whereas overestimating ω damps perturbation stress $\boldsymbol{\tau}_\Delta$ back toward linear RANS. Capping $\tau \leq 5.0$ s serves as an essential regularizer preventing singular invariant growth near the duct center where $\omega \to 0$.

10.3 Neural Network Model-Form Uncertainty

The 9.34% streamwise RMSE cannot be attributed solely to inlet boundary conditions. The neural network closure is an empirical approximation trained on frozen DNS gradients. In live CFD execution, the emerging secondary flow deforms the mean velocity field $\mathbf{U}$, modifying the velocity gradients fed back into the network at each SIMPLE iteration. This interactive feedback loop, combined with finite MLP expressivity and surrogate ω discrepancies, introduces genuine model-form uncertainty that must be transparently acknowledged.

11 Reproducibility & Open Source Verification Suite

The repository incorporates an automated test suite executed via `pytest` (`tests/`):

```
pytest -v
```

```
tests/test_coordinate_convention.py::test_non_square_gradient_ordering PASSED [ 5%]
tests/test_coordinate_convention.py::test_non_square_interpolation_ordering PASSED [ 10%]
tests/test_coordinate_convention.py::test_benchmark_dns_grid_convention PASSED [ 15%]
tests/test_mesh_reconstruction_verification.py::test_coarse_mesh_geometry_and_ordering PASSED [
     21%]
tests/test_mesh_reconstruction_verification.py::test_fine_mesh_geometry_and_ordering PASSED [
    26%]
tests/test_mesh_sensitivity.py::test_mesh_sensitivity_core_and_velocity PASSED [ 31%]
tests/test_parity.py::test_basis_trace_freedom PASSED [ 36%]
tests/test_parity.py::test_basis_symmetry PASSED [ 42%]
tests/test_parity.py::test_cardano_eigenvalue_accuracy PASSED [ 47%]
tests/test_parity.py::test_realizability_bounds PASSED [ 52%]
tests/test_parity.py::test_evaluator_parity PASSED [ 57%]
tests/test_parity.py::test_torchscript_parity PASSED [ 63%]
tests/test_parity.py::test_json_weights_parity PASSED [ 68%]
tests/test_parity.py::test_standalone_cpp_numerical_parity PASSED [ 73%]
tests/test_timescale_distribution.py::test_cfd_invariants_within_training_envelope PASSED [
    78%]
tests/test_timescale_distribution.py::test_basis_norms_bounded_by_training_distribution PASSED
    [ 84%]
tests/test_vortex_detection.py::test_zero_vorticity_field PASSED [ 89%]
tests/test_vortex_detection.py::test_known_four_vortex_field PASSED [ 94%]
tests/test_vortex_detection.py::test_known_eight_vortex_field PASSED [100%]

====================== 19 passed, 1 warning in 17.68s ======================
```

All raw DNS datasets, trained weights (`tbnn_model.pt`, `weights.json`), OpenFOAM solver sources, and automated evaluation scripts are available in the public repository:

https://github.com/BLACKNIGHT-cloud12/tbnnFoam

12 Conclusions & Future Work

In this study, we presented `tbnnFoam`, an open-source, in-situ coupled CFD solver integrating Tensor Basis Neural Networks into OpenFOAM for turbulence modeling beyond the linear Boussinesq hypothesis.

12.1 Summary of Findings

1. **Physical Anisotropy Recovery:** Live in-situ coupling successfully captures the nonlinear normal stress differences ($\overline{w'^2} - \overline{v'^2}$) that drive secondary motions of Prandtl's second kind, recovering all eight canonical secondary flow cells where conventional linear k–ω SST fails entirely.

2. **Dual Inference Parity:** Numerical equivalence between PyTorch, TorchScript C++, and a native zero-dependency C++ evaluator is demonstrated to $< 10^{-6}$ precision, with analytical Cardano cubic projection guaranteeing unconditional physical realizability.

3. **Discretization Sensitivity:** Increasing cross-stream resolution from 20×20 to 40×40 trends peak secondary circulation from 7.99% to 2.62% (approaching the 2.03% DNS benchmark) and reduces Q1 core localization error from $0.586\,h$ to $0.474\,h$ while preserving topological stability.

4. **Honest Evaluation of Discrepancies:** Streamwise velocity RMSE (9.34%) and secondary velocity overprediction are transparently characterized as a combination of inlet development length effects and neural network model-form discrepancy. Furthermore, joint 5D k-NN distance diagnostics reveal modest distribution shift ($0.21\,\sigma$ median) despite 100% marginal range coverage.

12.2 Future Directions

- **Differentiable CFD Training:** Transitioning from frozen-gradient a priori training to end-to-end differentiable RANS training (e.g., adjoint-coupled loss functions) to account directly for live velocity feedback during neural optimization.
- **Dynamic ω Formulation:** Training closures with explicit RANS-governed ω inputs to eliminate empirical surrogate assumptions.
- **Application to Complex Channels:** Evaluating the generalizability of `tbnnFoam` to curved passages, wavy ducts, and heat-exchanger subchannels.

References

[1] Ludwig Prandtl. Über die ausgebildete turbulenz. In *Proceedings of the 2nd International Congress for Applied Mechanics, Zurich*, pages 62–74, 1926.

[2] F B Gessner. The origin of secondary flow in turbulent flow along a corner. *Journal of Fluid Mechanics*, 58(1):1–25, 1973.

[3] E Brundrett and W D Baines. The production and diffusion of secondary circulation in turbulent flow through a square duct. *Journal of Fluid Mechanics*, 19(3):375–394, 1964.

[4] Charles G Speziale. Analytical methods for the development of reynolds-stress closures in turbulence. *Annual Review of Fluid Mechanics*, 23(1):107–157, 1991.

[5] T J Craft, B E Launder, and K Suga. Development and application of a cubic eddy-viscosity model of turbulence. *International Journal of Heat and Fluid Flow*, 17(2):108–115, 1996.

[6] Stefan Wallin and Arne V Johansson. An explicit algebraic reynolds stress model for incompressible and compressible turbulent flows. *Journal of Fluid Mechanics*, 403:89–132, 2000.

[7] Ricardo Vinuesa, Azad Noorani, Adrián Lozano-Durán, George K El Khoury, Philipp Schlatter, Paul F Fischer, and Hassan M Nagib. Aspect ratio effects in turbulent duct flows studied through direct numerical simulation. *Journal of Turbulence*, 15(10):677–706, 2014.

[8] Sergio Pirozzoli, Davide Modesti, Paolo Orlandi, and Francesco Grasso. Turbulence and secondary motions in square duct flow. *Journal of Fluid Mechanics*, 840:634–655, 2018.

[9] Julia Ling, Andrew Kurzawski, and Jeremy Templeton. Reynolds averaged turbulence modelling using deep neural networks with embedded invariance. *Journal of Fluid Mechanics*, 807:155–166, 2016.

[10] Stephen B Pope. A more general effective-viscosity hypothesis. *Journal of Fluid Mechanics*, 72(2):331–340, 1975.

[11] S Banerjee, R Krahl, F Durst, and C Zenger. Presentation of anisotropy properties of turbulence, invariants, versus eigenvalue approaches. *Journal of Turbulence*, 8:N32, 2007.

[12] Patrick J Roache. *Verification and Validation in Computational Science and Engineering*. Hermosa Publishers, Albuquerque, NM, 1998.

[13] ASME. *Standard for Verification and Validation in Computational Fluid Dynamics and Heat Transfer*. ASME V&V 20-2009. American Society of Mechanical Engineers, New York, NY, 2009.